

Lesson 4 Computational Thinking

Libraries

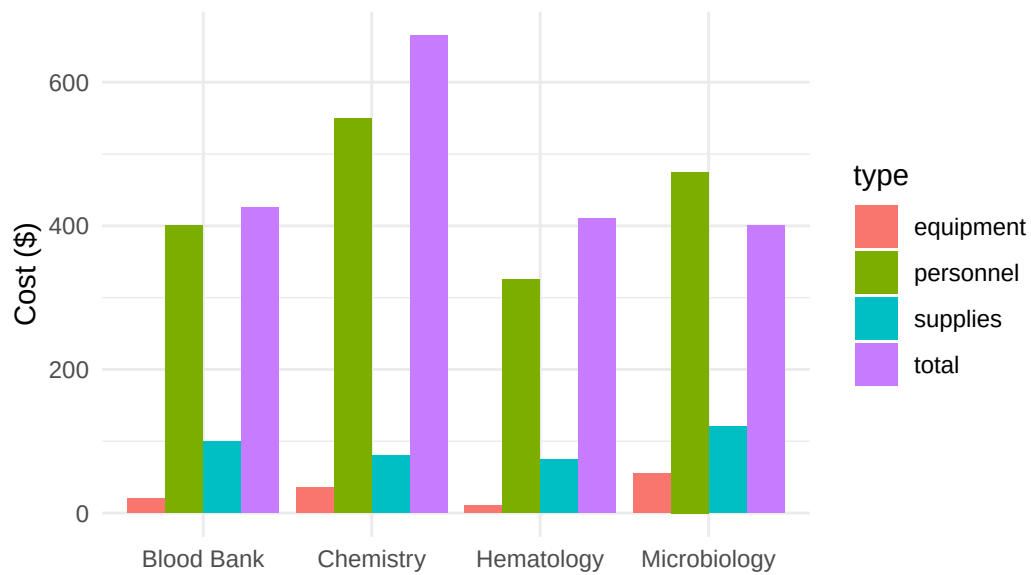
```
library(tidyverse)
```

Data import and preparation

```
exp <- read_csv("data/lab_exp.csv")
exp_long <- exp |> pivot_longer(personnel:total,
                                names_to = "type",
                                values_to = "cost")
```

Create graph

```
ggplot(data = filter(exp_long,
                      department %in% c("Chemistry", "Blood Bank",
                                         "Microbiology", "Hematology"))) +
  geom_bar(mapping = aes(x = department,
                        y = cost,
                        fill = type),
           stat = "identity",
           position = "dodge") +
  theme_minimal() +
  labs(x = "", y = "Cost ($)",
       title = "")
```



Conditionals

```
select_pan <- function(baked_good) {
  if (baked_good == "cupcake" | baked_good == "muffin") {
    pan <- "cupcake tin"
  } else if (baked_good == "cake") {
    pan <- "round cake pan"
  } else {
    pan <- "square cake pan"
  }
  return(pan)
}
```

```
select_pan(baked_good = "cupcake")
```

```
[1] "cupcake tin"
```

```
select_pan(baked_good = "muffin")
```

```
[1] "cupcake tin"
```

```
select_pan(baked_good = "cake")
```

```
[1] "round cake pan"
```

```
select_pan(baked_good = "brownie")
```

```
[1] "square cake pan"
```

Loops

```
# create a toy data set
df <- tibble(
  a = rnorm(10),
  b = rnorm(10),
  c = rnorm(10),
  d = rnorm(10)
)
```

```
# manually calculate median for each column via copy/paste
median(df$a)
```

```
[1] -0.4265819
```

```
median(df$b)
```

```
[1] -0.2323156
```

```
median(df$c)
```

```
[1] 0.268051
```

```
median(df$d)
```

```
[1] -0.3276386
```

```
# use a for loop to apply the median function across all columns

output <- vector("double", ncol(df)) # 1. output
for (i in seq_along(df)) {           # 2. sequence
  output[[i]] <- median(df[[i]])      # 3. body
}
output
```

```
[1] -0.4265819 -0.2323156  0.2680510 -0.3276386
```

```
# output is the result you want after executing code for each iteration in the sequence, here
# sequence defines what to loop over, here is each column of the dataframe
# body is the code that will be repeated, here it is the median function
```

```
# use a function that abstracts for loops
df |> map_dbl(median)
```

```
      a      b      c      d
-0.4265819 -0.2323156  0.2680510 -0.3276386
```

Prompting

```
glu <- read_csv("data/fasting_glucose_dataset.csv")
```

```
Rows: 300 Columns: 4
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (3): patient_id, site, sex
```

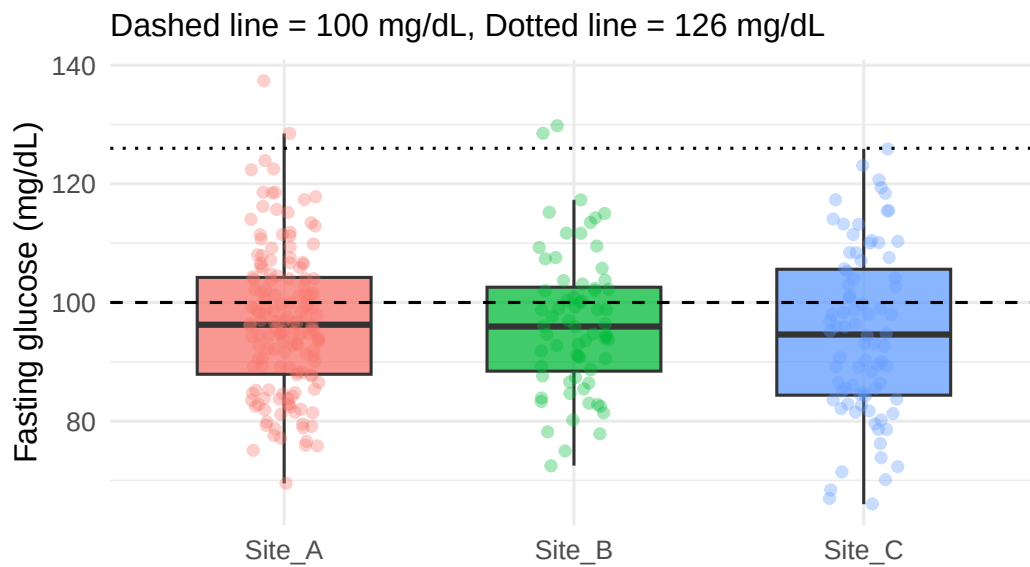
```
dbl (1): fasting_glucose_mg_dL
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ggplot(glu, aes(x = site, y = fasting_glucose_mg_dL, fill = site)) +
  geom_boxplot(width = 0.6, outlier.shape = NA, alpha = 0.75) +
  geom_jitter(aes(color = site), width = 0.12, alpha = 0.35, size = 1.6) +
  geom_hline(yintercept = 100, linetype = "dashed") +
  geom_hline(yintercept = 126, linetype = "dotted") +
```

```
labs(
  title = "",
  subtitle = "Dashed line = 100 mg/dL, Dotted line = 126 mg/dL",
  x = "",
  y = "Fasting glucose (mg/dL)"
) +
theme_minimal(base_size = 12) +
theme(legend.position = "none")
```



```
glu |> group_by(site) |> summarise(median = median(fasting_glucose_mg_dL))
```

```
# A tibble: 3 x 2
  site   median
  <chr>   <dbl>
1 Site_A  96.2
2 Site_B  96.0
3 Site_C  94.6
```

```
ggplot(glu, aes(x = site, y = fasting_glucose_mg_dL)) +
  geom_boxplot(fill = "white") +
  geom_jitter(width = 0.2, alpha = 0.6) +
```

```
labs(  
  title = "Fasting Glucose Distributions by Hospital Site",  
  y = "Fasting Glucose (mg/dL)",  
  x = NULL  
) +  
theme_minimal()
```

